(Please write your Enrollment Number)

Enrollment No._____

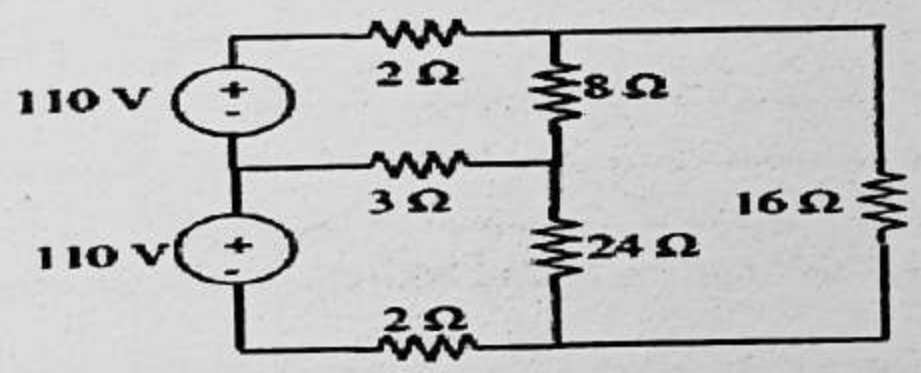
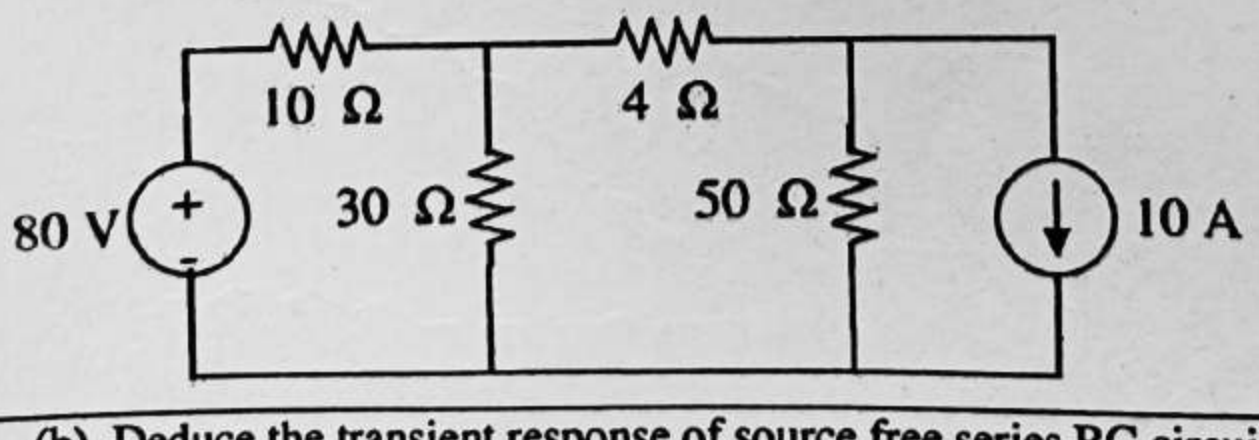
**Special End-Term Examination- ONLINE MODE**

**(CBCS)**

**B.Tech ECE IIIrd SEM**

**(Feb, 2022)**

| **Subject Code: BEC-205** | **Subject: Network analysis and synthesis** |
|---|---|
| **Time :** 1 Hour 15 Minutes | **Maximum Marks :** 30 |
| **Note:Q. 1 is compulsory. Attempt any one question from the rest.** | |

| | | |
|---|---|---|
| **Q1** | | **(5*3=15)** |
| | (a) Derive hybrid parameters in terms of Z parameter. | |
| | (b) How do you find Thevinin's resistance? | |
| | (c) A heater elements draws 20 A when connected to a 230-V line. How long does it take to consume 60 kJ? | |
| | | |
| | | |
| **Q2** | | **(7.5+7.5= 15)** |
| | (a) If two circuits X and Y are to be connected in cascade. Give the two port parameters of the combination with diagram of cascade combination. | |
| | (b) Using nodal analysis find all branch currents for the following circuit   | |
| | | |
| **Q3** | | **(7.5+7.5= 15)** |
| | (a) Verify Superposition theorem for 4Ω resistor for the following circuit.   | |
| | (b) Deduce the transient response of source free series RC circuit. | |